

Clock and Calendar

Calendar

- Ordinary year (O.Y.) = 365 days = 52 weeks + 1 day
- Leap year (L.Y.) = 366 days = 52 weeks + 2 days
- 100 years = 76 O.Y. + 24 L.Y.
- **Odd days :-** When total number of days is divided by 7, then the remainder is called odd days

In ordinary year = 1 odd day

In leap year = 2 odd days.

For 100 years = 76 O.Y. + 24 L.Y.

$$\begin{array}{ccc} \downarrow & & \downarrow \\ 76 \text{ odd days} & & 24 \times 2 = 48 \text{ odd days} \end{array}$$

$$\therefore \text{Total odd days in 100 years} = \frac{(76 + 48)}{7} = 124 / 7 = 5 \text{ odd days}$$

$$\therefore 100 \text{ years} = 5 \text{ odd days}$$

$$200 \text{ years} = (5 + 5 = 10) = 3 \text{ odd days}$$

$$300 \text{ years} (5 + 5 + 5 = 15) = 1 \text{ odd day}$$

$$400 \text{ years} = (5 + 5 + 5 + 6 = 21) = 0 \text{ odd day}$$

$\therefore$ (400 year is century year and also leap year so 1 day increases

- A year except century year divisible by 4 is called leap year.

- A century year divisible by 400 is called leap century.

No. of odd days in months.

- | | | | |
|-------------------|---|-------------------|-------------------|
| (1) January = 3 | (2) February = 1 (for L.Y)
= 0 (for O.Y) | (3) March = 3 | (4) April = 2 |
| (5) May = 3 | (6) June = 2 | (7) July = 3 | (8) August = 3 |
| (9) September = 2 | (10) October = 3 | (11) November = 2 | (12) December = 3 |

Week days code

- | | | | |
|--------------|------------|--------------|---------------|
| Sunday = 0 | Monday = 1 | Tuesday = 2 | Wednesday = 3 |
| Thursday = 4 | Friday = 5 | Saturday = 6 | |

Clock

Clock has two types of hand. Minute hand is called long hand while hour hand is known as short hand. In this topic there are some types of question such as :-

- (a) Questions on basic concepts.
- (b) Find out angle between both hands of a clock.
- (c) Find out accurate time for a given angle.

Concept : Clock works as a circle, such as it has complete 360° angle.

- (1) In an hour, a minute hand cover a complete circle or we can say 12 Blocks

$$12 \text{ Block} = 360^\circ$$

$$1 \text{ Block} = 30^\circ \quad [\because 1 \text{ Block} = 5 \text{ min}]$$

$$1 \text{ min} = 6^\circ$$

Note : The minute hand cover 6° angle within a minute.

- (2) In an hour, a hour hand covers 1 Block. For example. If its 4'0 clock and after one hour it will be at 5'0 clock.

$$1 \text{ hour} = 1 \text{ Block}$$

$$60 \text{ min} = 5 \text{ min} \quad [\because 1 \text{ Block} = 5 \text{ min}]$$

$$60 \text{ min} = 30^\circ, \therefore 1 \text{ min} = \frac{1}{2}^\circ$$

Note : An hour hand cover $\frac{1}{2}^\circ$ angle within a minute.

- Formula to find angle between hands of clock:-

$$\left| \frac{(11M - 60H)}{2} \right| \text{ or } \left| (M - 5H) \times 6 - \frac{M}{2} \right|$$

- When both the hands of a clock are in opposite direction to each other, when they are 30 minutes space apart and they make 180° . This situation occurs once in an hour, 11 times in 12 hours and 22 times in a day (24 hours) because in between 5 to 6 and 6 to 7, they are not in opposite direction.
- When the hands of the clock make 0° , they are said to be coincident or overlapping. This situation occurs once in an hour. 11 times in 12 hours and 22 times in a day (24 hours) because between 12 and 1, overlapping is not possible.
- When both the hands of clock are at right angle (when the distance between two hands measure 90°). At this time, they are 15 minutes space apart. This situation occurs twice in one hour, 22 times in 12 hours and 44 times in a day. In between 2 and 3, 3 and 4, one right angle is common and 8 to 9 only one times right angle will possible
- Between H and (H + 1) hours two hands coincide at = $\frac{60}{11} \times H$ past H

Solved Examples

1. At what angle, the hands of a clock are inclined at 40 minutes past 4 ? **Sol. (b); 10°**

- (a) 75° (b) 80°
(c) 90° (d) 100°

Sol. (d); 100°

$$\theta = \left| \frac{11M - 60H}{2} \right|$$

$$\theta = \left| \frac{11 \times 40 - 60 \times 4}{2} \right| = \left| \frac{200}{2} \right| = 100^\circ$$

2. An accurate clock shows 8 o'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 2 o'clock in the afternoon?

- (a) 144° (b) 150°
(c) 168° (d) 180°

Sol. (d); 180°

Angle traced by the hour hand in 6 hours

$$= \left(\frac{360}{12} \times 6 \right) = 180^\circ$$

3. The angle between the minute hand and the hour hand of a clock when the time is 4 : 20 is

- (a) 0° (b) 10°
(c) 5° (d) 20°

Angle traced by hour hand in

$$4 \frac{20}{60} = 4 \frac{1}{3} = \frac{13}{3} \text{ hrs is } \left(\frac{360^\circ}{12} \times \frac{13}{3} \right) = 130^\circ$$

Angle traced by minute hand in 20 min

$$= \left(\frac{360}{60} \times 20 \right) = 120^\circ$$

$\therefore$ Angle between the minute hand and hour hand of a clock when the time is 4:20

$$= 130^\circ - 120^\circ = 10^\circ$$

or

$$\theta = \left| \frac{11M - 60H}{2} \right| = \left| \frac{11 \times 20 - 60 \times 4}{2} \right|$$

$$= \left| \frac{220 - 240}{2} \right| = \left| \frac{-20}{2} \right| = | -10 |$$

$$\boxed{\theta = 10^\circ}$$

4. What was the day on 15th August 1947 ?

- (a) Friday (b) Monday
(c) Sunday (d) Wednesday

Sol. (a); Friday

1947 August 15



First of all make pair

$$1600 + 300 + 46 \text{ and } 15$$

$$\text{odd days in 1600 years} = 0$$

$$\text{odd days in 300 years} = 1$$

$$\text{odd days in 46 years} = 57$$

$$(\text{In 46 years} = 35 \text{ simple years} + 11 \text{ leap years})$$

$$\text{No of odd days in 35 simple years} = 35 \text{ odd days}$$

$$\text{No. of odd days in 11 leap years} = 2 \times 11 = 22 \text{ odd days}$$

$$\therefore 35 + 22 = 57 \text{ odd days}$$

Number of odd days between 1st January to 15 August

Jan	Feb	March	Apr.	May	June	July	Aug.
3	0	3	2	3	2	3	15

$$\therefore \text{Total no. of odd days} = 0 + 1 + 57 + 3 + 0 + 3 + 2 + 3 + 2 + 3 + 15 = 89$$

$$\begin{array}{r} 7 \overline{) 89} 12 \\ \underline{7} \\ 19 \\ \underline{14} \\ 5 \end{array}$$

Since, Remainder = 5

$\therefore$ It is Friday

5. If 6th march, 2005 is Monday, what was the day of the week on 6th march, 2004 ?

- (a) Sunday (b) Saturday
(c) Tuesday (d) Wednesday

Sol. (a); Sunday

$$2004 = \text{Leap year} = 2 \text{ odd days}$$

But, feb 2004 not included because we are calculating from March 2004 to March 2005, so it has 1 odd day

The day on 6th March, 2005 will be 1 day beyond the day on 6th March, 2004

Given that, 6th March, 2005 is Monday

$\therefore$ 6th March 2004 is Sunday (1 day before to 6th March 2005)

6. If the 1st October is Sunday. Then 1st November will be -

- (a) Tuesday (b) Wednesday
(c) Sunday (d) Friday

Sol. No of days between 1st Oct. to 1st Nov = 31

$$\begin{array}{r} 7 \overline{) 31} 4 \\ \underline{28} \\ 3 \end{array}$$

$$\therefore \text{Rem} = 3$$

$\therefore$ 3 days after Sunday will be Wednesday

Practice Set

- January 1, 2007 was Monday. What day of the week lies on Jan 1, 2008?
(a) Monday (b) Tuesday
(c) Wednesday (d) Sunday
- January 1, 2008 is Tuesday. What day of the week lies on Jan 1, 2009?
(a) Monday (b) Wednesday
(c) Thursday (d) Sunday
- On 8th Dec, 2007 Saturday falls. What day of the week was it on 8th Dec, 2006?
(a) Sunday (b) Thursday
(c) Tuesday (d) Friday
- On 6th March, 2005 Monday falls. What was the day of the week on 6th March 2004?
(a) Sunday (b) Saturday
(c) Tuesday (d) Wednesday
- The calendar for the year 2007 will be the same for the year?
(a) 2014 (b) 2016
(c) 2017 (d) 2018
- On what dates of April, 2001 did Wednesday fall?
(a) 1st, 8th, 15th, 22nd, 29th
(b) 2nd, 9th, 16th, 23rd, 30th
(c) 3rd, 10th, 17th, 24th
(d) 4th, 11th, 18th, 25th
- What was the day of the week on 17th June, 1998?
(a) Monday (b) Tuesday
(c) Wednesday (d) Thursday
- What was the day of the week on 28th May, 2006?
(a) Thursday (b) Friday
(c) Saturday (d) Sunday
- What will be the day of the week on 15th August, 2010?
(a) Sunday (b) Monday
(c) Tuesday (d) Friday
- Today is Monday. After 61 days, it will be :
(a) Wednesday (b) Saturday
(c) Tuesday (d) Thursday

11. The last day of a century cannot be :
 (a) Monday (b) Wednesday
 (c) Tuesday (d) Friday
12. Which of the following is not a leap year?
 (a) 700 (b) 800
 (c) 1200 (d) 2000
13. How many days are there in x weeks x days?
 (a) 7×2 (b) $8x$
 (c) $14x$ (d) 7
14. It was Sunday on Jan 1, 2006. Find the day of the week on Jan 1, 2010?
 (a) Sunday (b) Saturday
 (c) Friday (d) Wednesday
15. On 8th Feb, 2005 it was Tuesday. What was the day of the week on 8th Feb, 2004?
 (a) Tuesday (b) Monday
 (c) Sunday (d) Wednesday
16. An accurate clock shows 9 o'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 2 o'clock in the afternoon?
 (a) 144° (b) 150°
 (c) 168° (d) 180°
17. How many times are the hands of a clock at right angle in a day?
 (a) 22 (b) 24
 (c) 44 (d) 48
18. How many times do the hands of a clock coincide in a day?
 (a) 22 (b) 24
 (c) 44 (d) 48
19. What is the angle between the hands of clock at 3:15AM?
 (a) 10° (b) 0°
 (c) 7.5° (d) 15°
20. What is the angle between hands of a clock at 4:15 AM
 (a) 37.5° (b) 40.5°
 (c) 30° (d) 32°
21. A clock is started at noon. By 10 minutes past 5, the hour hand has turned through:
 (a) 145° (b) 150°
 (c) 155° (d) 160°
22. How many times the hands of clock will be opposite direction in a day?
 (a) 24 (b) 23
 (c) 22 (d) 21
23. The angle between the minute hand and the hour hand of a clock when the time is 8.30, is:
 (a) 80° (b) 75°
 (c) 60° (d) 105°
24. How many times in a day, are the hands of a clock in straight line but opposite in direction?
 (a) 22 (b) 24
 (c) 44 (d) 48
25. The angle between the minute hand and the hour hand of a clock when the time is 4.30, is:
 (a) 0° (b) 45°
 (c) 5° (d) 20°
26. Which of the following is a leap year ?
 (a) 900 (b) 600
 (c) 1600 (d) 1800
27. 9 days before, mohani went to watch cinema. She goes to watch cinema only on Thursday. What day of week, is today?
 (a) Saturday (b) Thursday
 (c) Sunday (d) Friday
28. If Jan 1, 2011 was Saturday, then what day was on 31 dec, 2011?
 (a) Friday (b) Saturday
 (c) Sunday (d) Tuesday
29. If Jan 1, 2012 was Sunday then what day was on Jan 1, 2013 ?
 (a) Sunday (b) Monday
 (c) Tuesday (d) Wednesday
30. Which of the following can be first day of any century?
 (a) Friday (b) Wednesday
 (c) Monday (d) None of these
31. Which of the following can be last day of any century?
 (a) Saturday (b) Thursday
 (c) Friday (d) Tuesday
32. What day was on Jan 1 AD
 (a) Monday (b) Tuesday
 (c) Sunday (d) Friday
33. Rahim was born on 8th April, 1985. It was Saturday. On which day will he be 15 years, 6 months and 10 days?
 (a) Sunday (b) Saturday
 (c) Monday (d) Tuesday
34. If the day, two days after tomorrow is a Sunday. What will be the day, three days before yesterday?
 (a) Sunday (b) Saturday
 (c) Friday (d) Monday
35. If 15th March, 1999 was a Monday. What was the day the 10th July, 1999?
 (a) Friday (b) Tuesday
 (c) Saturday (d) Wednesday
36. At what angle, the hands of a clock are inclined at 42 minutes past 5?
 (a) 90° (b) 80°
 (c) 78° (d) 81°

37. At what angle, the hands of a clock are inclined at 22 minutes past 9?
 (a) 145° (b) 149°
 (c) 85° (d) 115°
38. At what angle, the hands of a clock are inclined at 15 minutes past 5?
 (a) 86.5° (b) 90°
 (c) 78° (d) 67.5°
39. At what time between 6 and 7, are the hands of a clock coincides?
 (a) 32 minutes past 6
 (b) 30 minutes past 6
 (c) $32\frac{8}{11}$ minutes past 6
 (d) None of these
40. A clock is set right at 12:00 noon. The clock loses 20 minutes in 24 hours. How many minute the clock will lose till 6 am next day?
 (a) 12 min (b) 20 min
 (c) 18 min (d) 15 min
41. If day before yesterday is a Saturday. What will be sixth day after tomorrow?
 (a) Sunday (b) Saturday
 (c) Monday (d) Tuesday
42. Looking into a mirror, the clock shows 3:35 as the time. The actual time is--
 (a) 8:25 (b) 9:25
 (c) 7:35 (d) 8:35
43. If tomorrow is Tuesday. What day will be day before three days from the day after tomorrow?
 (a) Sunday (b) Monday
 (c) Tuesday (d) Wednesday
44. If the real time is 14 : 25 in a clock. What time a clock will show when we are looking in mirror?
 (a) 8:35 (b) 9:25
 (c) 9:35 (d) 8:55
45. Find the angle between the hands of clock at 01 : 15 AM
 (a) 60° (b) 23°
 (c) 52.5° (d) 62.5°
46. Find the angle between the hands of clock at 12 : 30 ?
 (a) 165° (b) 170°
 (c) 175° (d) 155°
47. If the day before yesterday was Wednesday. What day will be next from day after tomorrow?
 (a) Sunday (b) Monday
 (c) Tuesday (d) Wednesday
48. Six bells rings at regular intervals of 2sec, 4sec, 6sec, 8sec, 10sec, 12sec. How many times will they ring together in 30 minutes?
 (a) 16 times (b) 17 times
 (c) 15 times (d) None of these
49. If the actual time is 11 : 30 and when we move minute hand towards clockwise direction of 135° angle. Then what will be time in clock?
 (a) 52 minute past 11 (b) $52\frac{1}{2}$ minute past 11
 (c) 51 minute past 11 (d) 50 minute past 11
50. If a clock is showing 4 am. Then we move the hour hand towards clockwise direction 75° angle then what will be time in clock?
 (a) 7 : 25 (b) 7 : 40
 (c) 6 : 30 (d) 7 : 50

Distinct Questions

51. A watch which gains uniformly is 2 minutes low at noon on Monday and is 4 min. 48 sec fast at 2 p.m. on the following Monday. When was it correct?
 (a) 2 p.m. on Tuesday
 (b) 2 p.m. on Wednesday
 (c) 3 p.m. on Thursday
 (d) 1 p.m. on Friday
52. At what time, in minutes, between 3 o'clock and 4 o'clock, both the needles will coincide each other?
 (a) $5\frac{1}{11}$ min past 3 (b) $12\frac{4}{11}$ min past 3
 (c) $13\frac{4}{11}$ min past 3 (d) $16\frac{4}{11}$ min past 3
53. The train for Delhi from Bhiwani leaves every $4\frac{1}{2}$ hours. The enquiry clerk told a passenger that the train for Delhi left 40 minutes ago and next train will be leave at 7.20 pm. what was the time when enquiry clerk told this?
 (a) 3 : 30 (b) 2 : 50
 (c) 4 : 00 (d) 6 : 20
54. A national leader celebrated his birthday on 29th Feb. Now, after how many years he will celebrate his birthday?
 (a) 2 years (b) 3 years
 (c) 4 years (d) None of these
55. Ram was asking bus from Delhi to Mumbai. His clock is 10 minutes fast from right time. According to his clock, it was 12 : 20 pm. A traveller told that bus for Delhi to Mumbai leaves every 55 minutes and bus left 25 minutes ago. Then what is the right time of next bus?
 (a) 1 : 00 pm (b) 5 : 00 pm
 (c) 12 : 40 pm (d) None of these

56. If any year on 3rd March, it was Tuesday then what day on the Gandhi Jayanti in the same year?
 (a) Saturday (b) Friday
 (c) Monday (d) None of these
57. In 1996, the republic day was celebrated on Saturday. In the same year on which day the independence day was celebrated?
 (a) Friday (b) Saturday
 (c) Sunday (d) Tuesday
58. A watch which runs uniformly, is 4 minutes slow at 3am on Tuesday and 2 minutes fast at 3 am on. Thursday. When was it correct?
 (a) 11 am on Wednesday
 (b) 6 am on Thursday
 (c) 11 pm on Wednesday
 (d) 11 pm on Tuesday
59. If the actual time is 12:40, Then what time a clock will show when we are looking in a mirror?
 (a) 11:20 (b) 11:30
 (c) 10:20 (d) 9:40
60. Five bells rings at regular intervals of 2 sec, 3sec, 4sec, 5sec and 6 sec. How many times will they ring together in an hour?
 (a) 60 times (b) 62 times
 (c) 63 times (d) 61 times

Previous year Questions (Memory Based)

1. A clock is set right at 8 am. The clock gains 10 minute in 24 hours. What will be the true time when the clock indicates 1 p.m. on the following day?
 (a) 48 min. past 12 (b) 46 min. past 12
 (c) 45 min. past 12 (d) 47 min. past 12
2. The calendar for the year 1993 will be same for the year.
 (a) 2004 (b) 1999
 (c) 1998 (d) 2003
3. A clock is set right at 5 a.m. The clock loses 16 minutes in 24 hours. What will be the true time when the clock indicates 10 pm. on 4th day.
 (a) 11 pm (b) 12 pm
 (c) 1 pm (d) 2 pm
4. What was the day of the week on. 16th July 1776
 (a) Tuesday (b) Wednesday
 (c) Monday (d) Saturday
5. How many days are there in X weeks X days.
 (a) $7x^2$ (b) $8x$
 (c) $14x$ (d) $7x$
6. If immediate next day of Tomorrow is Sunday then what will be 3rd day before yesterday.
 (a) Monday (b) Tuesday
 (c) Saturday (d) Friday
7. A watch which gains uniformly is 5 min, slow at 8 O' clock in the morning on Sunday and it is 5 min 48 sec. Fast at 8 pm. on following Sunday. When was it correct
 (a) 7 pm on Wednesday
 (b) 20min past 7pm on Wednesday
 (c) 18 min past 7 pm on Wednesday
 (d) 8 pm on Wednesday
8. At what time between 5 and 6 will the hands of a clock coincide:
 (a) 200/11 (b) 300/11
 (c) 400/11 (d) 250/11
9. In a whole 12 hours the minute hand overlap to the hour hand.
 (a) 11 times (b) 23 times
 (c) 22 times (d) 24 times
10. What times in a day or 24 hours the angle between minute hand and hour hand is right angle.
 (a) 44 times (b) 22 times
 (c) 11 times (d) 12 times
11. What time between 4 and 5 O'clock will the hands of a clock be at rightangle for 2nd time.
 (a) $3\frac{8}{11}$ (b) $38\frac{2}{11}$
 (c) $45\frac{9}{11}$ (d) $40\frac{1}{11}$
12. Find out the angle between hour hand and minute hand at 6 : 49.
 (a) 89.5° (b) 90°
 (c) 82° (d) 110°
13. Find out the reflexive angles at 11:05.
 (a) 57.5° (b) 360.5°
 (c) 302.5° (d) 190°
14. At what angle the hands of a clock are undivided at 15 minutes past 5.
 (a) 57.5° (b) 67.5°
 (c) 77.5° (d) 87.5°
15. In every 60 minutes, the minute hand gains minutes on the hour hand.
 (a) 53 (b) 54
 (c) 55 (d) 56
16. At what time, in minutes, between 3 O' clock and 4 O' clock both the needles will coincide each other.
 (a) $11\frac{4}{11}$ past 4 (b) $13\frac{4}{11}$ past 3
 (c) $15\frac{4}{11}$ past 3 (d) $16\frac{4}{11}$ past 3

17. An accurate clock shows 8 O' clock in the morning. Through how many degrees will the hour hand rotate when the clock show 2 O' clock in the afternoon.
 (a) 360° (b) 180°
 (c) 90° (d) 60°
18. The angle between minute hand and hour hand at 10 :20.
 (a) 170° (b) 200°
 (c) 120° (d) 150°
19. How much does a watch lose per day, if its hands coincide every 64 minutes.
 (a) $32\frac{8}{11}$ (b) $33\frac{8}{11}$
 (c) $34\frac{8}{11}$ (d) $35\frac{8}{11}$
20. Which year's calender is repeating calender for year 2016.
 (a) 2024 (b) 2023
 (c) 2044 (d) 2045
21. If in any month on third day it is Tuesday. How many maximum Sunday can be in the month?
 (a) 4 (b) 5
 (c) 3 (d) None of these
22. If on August 2nd, is Friday. What day will be day after thirteen day from 12 August?
 (a) Monday (b) Sunday
 (c) Wednesday (d) Saturday
23. In a 30 days month, there are five Saturday. What will be the first day of month?
 (a) Saturday (b) Friday
 (c) Friday or Saturday (d) Sunday
24. In any special month, three Sunday are on even number date which day will be on 15th date of month?
 (a) Thursday (b) Friday
 (c) Saturday (d) Sunday
25. If 3rd January is Sunday. Which date will be after three days from fourth Wednesday?
 (a) 30 January (b) 27 January
 (c) 26 January (d) 23 January
26. If in any month, 5th day is after two days from Monday. What day will be 18th in the month?
 (a) Monday (b) Tuesday
 (c) Wednesday (d) Thursday
27. If in December month on 17th day is Monday. What day will be maximum times in the month?
 (a) Monday (b) Tuesday
 (c) Wednesday (d) Friday
28. If on October 25, is Thursday, How many Monday will be in October month?
 (a) 3 (b) 4
 (c) 5 (d) 6
29. How many times the hands of clock will make right angle in a day?
 (a) 48 (b) 44
 (c) 43 (d) 42
30. In any year, world environment day was celebrated on Friday. In the same year on which day the children day was celebrated?
 (a) Friday (b) Tuesday
 (c) Saturday (d) Wednesday

Practice Set Solutions

1. **(b);** The year 2007 is an ordinary year. So, it has 1 odd day.
 1st day of the year 2007 was Monday.
 1st day of the year 2008 will be 1 day beyond Monday.
 Hence, it will be Tuesday.
2. **(c);** The year 2008 is a leap year. So, it has 2 odd days.
 1st day of the year 2008 is Tuesday (Given)
 So, 1st day of the year 2009 is 2 days beyond Tuesday.
 Hence, it will be Thursday.
3. **(d);** The year 2006 is an ordinary year. So, it has 1 odd day.
 So, the day on 8th Dec, 2007 will be 1 day beyond the day on 8th Dec, 2006.
 But, 8th Dec, 2007 is Saturday.
 $\therefore$ 8th Dec, 2006 is Friday.
4. **(a);** The year 2004 is a leap year. So, it has 2 odd days.
 But, Feb 2004 not included because we are calculating from March 2004 to March 2005. So it has 1 odd day only.
 The day on 6th March, 2005 will be 1 day beyond the day on 6th March, 2004.
 Given that, 6th March, 2005 is Monday.
 6th March, 2004 is Sunday (1 day before to 6th March, 2005).
5. **(d);** Count the number of odd days from the year 2007 onwards to get the sum equal to 0 odd day.
Year : 2007 2008 2009 2010 2011 2012 2013
 2014 2015 2016 2017
Odd day : 1 2 1 1 1 2 1 1 1 2 1
 Sum = 14 odd days 0 odd days.
 Calendar for the year 2018 will be the same as for the year 2007.

6. (d); We shall find the day on 1st April, 2001.
 1st April, 2001 = (2000 years + Period from 1.1.2001 to 1.4.2001)
 Odd days in 1600 years = 0
 Odd days in 400 years = 0
 Jan. Feb. March April
 $(31 + 28 + 31 + 1) = 91$ days 0 odd days.
 Total number of odd days = $(0 + 0 + 0) = 0$
 On 1st April, 2001 it was Sunday.
 In April, 2001 Wednesday falls on 4th, 11th, 18th and 25th.
7. (c); 17th June, 1998 = (1997 years + Period from 1.1.1998 to 17.6.1998)
 Odd days in 1600 years = 0
 Odd days in 300 years = $(5 \times 3) = 1$ odd day
 97 years has 24 leap years + 73 ordinary years.
 Number of odd days in 97 years $(24 \times 2 + 73) = 121 = 2$ odd days.
 Jan. Feb. March April May June
 $(31 + 28 + 31 + 30 + 31 + 17) = 168$ days
 168 days = 24 weeks = 0 odd day.
 Total number of odd days = $(0 + 1 + 2 + 0) = 3$.
 Given day is Wednesday.
8. (d); 28 May, 2006 = (2005 years + Period from 1.1.2006 to 28.5.2006)
 Odd days in 1600 years = 0
 Odd days in 400 years = 0
 5 years = (4 ordinary years + 1 leap year) = $(4 \times 1 + 1 \times 2) = 6$ odd days
 Jan. Feb. March April May
 $(31 + 28 + 31 + 30 + 28) = 148$ days
 148 days = (21 weeks + 1 day) 1 odd day.
 Total number of odd days
 $= (0 + 0 + 6 + 1) = 7$ 0 odd day.
 Given day is Sunday.
9. (a); 15th August, 2010 = (2009 years + Period 1.1.2010 to 15.8.2010)
 Odd days in 1600 years = 0
 Odd days in 400 years = 0
 9 years = (2 leap years + 7 ordinary years) = $(2 \times 2 + 7 \times 1) = 11 = 4$ odd days.
 Jan. Feb. March April May June July Aug.
 $(31 + 28 + 31 + 30 + 31 + 30 + 31 + 15) = 227$ days
 227 days = (32 weeks + 3 days) 3 odd days.
 Total number of odd days = $(0 + 0 + 4 + 3) = 7 = 0$ odd days.
 Given day is Sunday.
10. (b); Each day of the week is repeated after 7 days.
 So, after 63 days, it will be Monday.
 After 61 days, it will be Saturday.
11. (c); 100 years contain 5 odd days.
 Last day of 1st century is Friday.
 200 years contain $(5 \times 2) = 3$ odd days.
 Last day of 2nd century is Wednesday.
 300 years contain $(5 \times 3) = 15 = 1$ odd day.
 Last day of 3rd century is Monday.
 400 years contain 0 odd day.
 Last day of 4th century is Sunday.
 This cycle is repeated.
 Last day of a century cannot be Tuesday or Thursday or Saturday.
12. (a); The century divisible by 400 is a leap year.
 The year 700 is not a leap year.
13. (b); x weeks $\times$ days = $(7x + x)$ days = $8x$ days.
14. (c); On 31st December, 2005 it was Saturday.
 Number of odd days from the year 2006 to the year 2009 = $(1 + 1 + 2 + 1) = 5$ days.
 On 31st December 2009, it was Thursday.
 Thus, on 1st Jan, 2010 it is Friday.
15. (c); The year 2004 is a leap year. It has 2 odd days.
 The day on 8th Feb, 2004 is 2 days before the day on 8th Feb, 2005.
 Hence, this day is Sunday.
16. (b); Angle traced by the hour hand in 5 hours = $(360 / 12 \times 5) = 150^\circ$.
17. (c); In 12 hours, they are at right angles 22 times.
 In 24 hours, they are at right angles 44 times.
18. (a); The hands of a clock coincide 11 times in every 12 hours (Since between 11 and 1, they coincide only once, i.e., at 12 o'clock).
AM: 12:00, 1:05, 2:11, 3:16, 4:22, 5:27, 6:33, 7:38, 8:44, 9:49, 10:55
PM: 12:00, 1:05, 2:11, 3:16, 4:22, 5:27, 6:33, 7:38, 8:44, 9:49, 10:55
 The hands coincide 22 times in a day.
19. (c); $15 \times \frac{1}{2} = 7.5^\circ$
20. (a); $30^\circ + \frac{15}{2} = 37.5^\circ$
21. (c); Angle traced by hour hand in 12 hrs = 360° .
 Angle traced by hour hand in 5 hrs 10 min i.e.,
 $31/6$ hrs = $\left(\frac{360}{12} \times \frac{31}{6} \right) = 155^\circ$
22. (c); 22 times the hands of clock will be opposite directions in a day.

23. (b); Angle traced by hour hand in $\frac{17}{2}$ hrs

$$= \left(\frac{360}{12} \times \frac{17}{2} \right)^\circ$$

Angle traced by min. hand in 30 min is

$$\left(\frac{360}{60} \times 30 \right)^\circ = 180^\circ$$

Required angle = $(255 - 180)^\circ = 75^\circ$.

24. (a); The hands of a clock point in opposite directions (in the same straight line) 11 times in every 12 hours. (Because between 5 and 7 they point in opposite directions at 6 o'clock only).

So, in a day, the hands point in the opposite directions 22 times.

25. (b); Angle traced by hour hand in $(9/2)$ hrs

$$= \left(\frac{360}{12} \times \frac{9}{2} \right)^\circ = 135^\circ$$

Angle traced by min. hand in 30 min.

$$= \left(\frac{360}{60} \times 30 \right)^\circ = 180^\circ$$

Required angle = $(180 - 135)^\circ = 45^\circ$.

26. (c); All are century years divisible by 400 will be leap year. 1600 is divisible by 400. So 1600 is a leap year.

27. (a); Thursday + 9 days = Saturday

28. (b); In ordinary year 1 Jan. and 31 Dec. will be same day. If Jan first is Saturday then 31 Dec. will be Saturday also.

29. (c); In leap year 1 Jan. is Sunday then last day or 31 Dec. will be next day from 1 Jan. If here Jan 1, is Sunday then 31st Dec. will be Monday. And on 1 Jan 2013 will be next day means Tuesday.

30. (c); 31. (c);

32. (a); It was Monday on Jan 1, A.D.

33. (c); 8th April 1985 + (diff. between years + no. of leap year + no. of odd days from Jan. 1 to October 18)
 $= \text{Saturday} + (15 + 3 + 3 + 1 + 3 + 2 + 3 + 2 + 3 + 3 + 2 + 18) \div 7$
 $= \text{Saturday} + 58 \div 7$
 $= \text{Saturday} + 2 \text{ days} = \text{Monday}$

34. (a); I I I Yesterday Today tomorrow I I
 $\downarrow$ Sunday
Sunday

35. (c); Day between 15th March to 10 July.

March = 16

April = 30

May = 31

June = 30

July $\frac{= 10}{= 117}$

Hence, no. of odd days = 5

$\therefore$ Monday + 5 = Saturday

36. (d); Required time =

$$= (42 - 5 \times 5) \times 6 - \frac{42}{2} = (42 - 25) \times 6 - 21$$

$$17 \times 6 - 21 = 102^\circ - 21^\circ = 81^\circ$$

37. (b); Required time = $(22 - 5 \times 9) \times 6 - \frac{22}{2}$

$$(22 - 45) \times 6 - 11 = (-23) \times 6 - 11$$

$$= -138 - 11 = -149 = 149^\circ$$

38. (d); Required time = $(15 - 5 \times 5) \times 6 - \frac{15}{2}$

$$(15 - 25) \times 6 - 7.5 = -10 \times 6 - 7.5$$

$$= -60 - 7.5 = -67.5 = 67.5^\circ$$

39. (c); Required time = $\frac{60}{11} \times (\text{Time} \pm 0)$ past 6

$$= \frac{60}{11} \times (6 + 0) = \frac{60}{11} \times 6 = \frac{360}{11} = 32 \frac{8}{11} \text{ past 6}$$

40. (d); Time from 12 : 00 to 6 : 00 am = 18 hrs.

$\therefore$ in 24 hours clock loses = 20 min

$\therefore$ In 1 hour clock loses = $\frac{20}{24}$ min

$\therefore$ In 18 hours clock loses $\frac{20}{24} \times 18 = 15$ minutes.

41. (c); I Yesterday Today tomorrow +1 + 5
 $\downarrow$ Tuesday Saturday
i.e. Tuesday + 6 $\rightarrow$ Monday

42. (a); 11 : 60 [$\because$ For every Question, based
- 3 : 35 on mirror we should subtract
8 : 25 actual time from 11 : 60 (if the
time is in between 1 to 11)]

43. (a); I Today tomorrow 1
 $\downarrow$
Sunday Tuesday 1

44. (c); 23 : 60
- 14 : 25
9 : 35

45. (c); Angle between hands of clock = $60^\circ - 15 \times \frac{1}{2}$
 $= 52.5^\circ$

46. (a); Angle between hands of clock = $180 - 30 \times \frac{1}{2}$
 $= 165^\circ$

47. (b);

1	yesterday	Today	tomorrow	1	1
↓				↓	↓
Wednesday	→				Monday

48. (a);

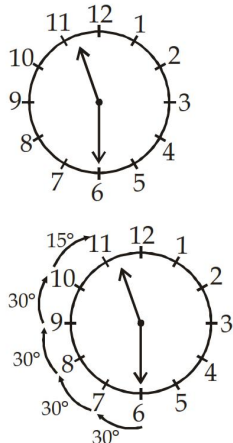
2	2, 4, 6, 8, 10, 12
2	1, 2, 3, 4, 5, 6
3	1, 1, 3, 2, 5, 3
	1, 1, 1, 2, 5, 1

$2 \times 2 \times 3 \times 2 \times 5 = 120$ second = 2 minutes

∴ Bells ring together 1 times in 2 mnutes.

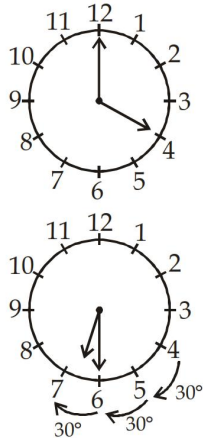
In 30 minutes bells ring together = $15 + 1 = 16$ times.

49. (b);



Clearly, time will be $11 : 52 \frac{1}{2}$ O'clock.

50. (c);



Clearly, time will be 6 : 30

Distinct Solutions

51. (b); Time from 12 p.m. on Monday to 2 p.m. on the following Monday = 7 days 2 hours = 170 hours.

The watch gains $(2 + 4\frac{4}{5})$ min or $\frac{34}{5}$ min in 170 hrs

Now, $\frac{34}{5}$ min. are gained in 170 hrs.

2 min. are gained in $(170 \times \frac{5}{34} \times 2)$ hrs = 50 hrs.

Watch is correct 2 days 2 hrs. after 12 p.m. on Monday i.e., it will be correct at 2 p.m. on Wednesday.

52. (d); At 3 o'clock, the minute hand is 15 min. spaces apart from the hour hand.

To be coincident, it must gain 15 min. spaces. 55 min. are gained in 60 min.

15 min. are gained in $(60/55 \times 15)$ min = $16\frac{4}{11}$ min

The hands are coincident at $16\frac{4}{11}$ min past 3

53. (a); Next train time → 7 : 20

Time of previous train → 7 : 20 - 4 : 30
 $= 2 : 50$ pm = 2 : 50 + 40 min = 3 : 30 pm

54. (c); We know that feb 29 only in leap year. Leap year comes again after 4 years.

55. (c); Present right time is 12 : 20 - 0 : 10
 $= 12 : 10$ pm

Previous bus time was 12 : 10 - 0 : 25 = 11 : 45 am
 So, time of next bus = 11 : 45 + 0 : 55 = 12 : 40 pm.

56 (b); Days between 3rd March to 2nd Oct.

March = 28	April = 30
May = 31	June = 30
July = 31	Aug. = 31

Sept. = 30	Oct. = $\frac{2}{213}$
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No. of odd days = $213/7 = 3$

∴ Tuesday + 3 days = Friday

57. (a); Jan. = 5

Feb. = 29

March = 31

April = 30

May = 31

June = 30

July = 31

Aug. $\frac{15}{202}$

∴ No. of odd days = 6

∴ Saturday + 6 days = Friday

58. (a); Time from 3 am Tuesday to 3 am Thursday = 48 hours.

In 48 hours clock runs fast = 6 minutes

$\therefore$ Clock 6 minutes runs fast = 48 hours

$\therefore$ Clock 1 minute runs fast = $\frac{48}{6}$ hours.

$\therefore$ Clock 4 minutes runs fast $\frac{48}{6} \times 4 = 32$ hours

Tuesday + 32 hours = on Wednesday 11 : 00 am
($\therefore$ clock will show right time when it will run 4 minutes fast)

59. (a);
$$\begin{array}{r} 23 : 60 \\ - 12 : 40 \\ \hline 11 : 20 \end{array}$$

60. (d);
$$\begin{array}{c|cccc} 2 & 2, & 3, & 4, & 5, & 6 \\ \hline 3 & 1, & 3, & 2, & 5, & 3 \\ \hline & 1, & 1, & 2, & 5, & 1 \\ \hline \end{array}$$

$$2 \times 3 \times 2 \times 5 = 60 \text{ second} = 1 \text{ minute}$$

bells ring 1 time in 1 min

$\therefore$ So in 60 minutes, the bells ring together
60 times + 1 = 61 times

($\therefore$ Here in such kind of questions, we should add one because when the bells ring. First time together we count 1)

Previous year Solutions (Memory Based)

1. (a); Time from 8 a.m. on a day to 1 p.m. on the following day = 29 hrs.
24 hours 10 min. of this clock = 24 hours of the correct clock.

$$\frac{145}{6} \text{ hrs of this clock} = 24 \text{ hours of correct clock}$$

$$29 \text{ hours of this clock} = \frac{24 \times 6 \times 29}{145} \text{ hrs of correct clock}$$

= 28 hrs 48 in of correct clock.

Therefore, the correct time is 28 hrs 48 min after 8 a.m. That is 48 min. past 12.

2. (b); Given year is a general year then add digit 6
 $1993 + 6 = 1999$, is same as 1993.
3. (a); Time from 5 a.m. on a day to 10 p.m. on 4th day = 89 hours.
Now 23 hrs 44 min of this clock = 24 hours of correct clock.
= $356/15$ hrs of the clock = 24 hours of correct clock.

$$= 89 \text{ hrs of this clock} = \frac{24 \times 15 \times 89}{356} \text{ hrs of correct clock.}$$

clock.

= 90 hrs of correct clock.

So, the correct time is 11 Pm.

4. (a); 16th July 1776 = (1775 years + period from 1776 to 16th July 1776), counting of odd days :
1600 years have 0 odd day.
100 years have 5 odd days.
75 years = (18 leap years + 57 ordinary years)
= $(18 \times 2 + 57 \times 1) = 93$ odd days
= 13 week + 2 days = 2 odd days.

1775 years have odd days = $(0 + 5 + 2) = 7 = 0$ odd day

Jan to 16th July extra day = 28 week + 2 days
= 2 days

So total no. of odd days = 2 days

So the day is Tuesday.

5. (b); Total days = $7x + x = 8x$ days
6. (a); If next day of tomorrow is Sunday then difference between 3rd day before yesterday is 6.
So the day will be Sunday - 6 = Monday.
7. (b); This Sunday morning at 8 : 00 am., the watch is 5 min slow and the next Sunday at 8.00 pm it become 5 min 48 sec fast. The watch gains

$$5 + 5 \frac{48}{60} = \frac{54}{5} \text{ min in a time of } (7 \times 24) + 12 = 180 \text{ hours.}$$

To show the correct time, it has to gain 5 min.

$$\frac{54}{5} \text{ min} = 180 \text{ hours, } 5 \text{ min} = \frac{180}{\frac{54}{5}} \times 5$$

$$= 83 \frac{1}{3} \text{ hrs}$$

$$\text{Total hrs.} = 72 \text{ hrs} + 11 \frac{1}{3} \text{ hrs}$$

$$= 3 \text{ day} + 11 \text{ hrs} + 20 \text{ min.}$$

$\therefore$ 20 min. past 7 pm on wednesday

8. (b); Let at x minutes hands coincide

$$6x - \left(\frac{x}{2} + 150 \right) = 0$$

$$x = \frac{300}{11} \text{ min}$$

9. (a); The both hands overlap to each other in an hour 1 times. In 12 hours it will be 11 times.
10. (a); Right angle between minute hand and hour hand occur two times in a hours.
12 hours = 22 times
24 hours = 44 times.
11. (b); Between 4 and 5' O clock, there are two times right angles occur. First at 5 min, second one at 7 block means 35 min so

$$4 \text{ O' clock } 35 \times \frac{12}{11} = \frac{420}{11} = 38 \frac{2}{11}$$

12. (a); Angle between hour hand and minute hand.

$$= \left| \frac{11 \times 49 - 60 \times 6}{2} \right| = \left| \frac{179}{2} \right| = 89.5^\circ$$

13. (c); Angle between both hands at 11.05 O' clock

$$= \left| \frac{11 \times 5 - 60 \times 11}{2} \right| = \left| \frac{-605}{2} \right| = 302.5^\circ$$

14. (b); Angle between both hand at 5.15

$$= \left| \frac{11 \times 15 - 60 \times 5}{2} \right| = \left| \frac{-135}{2} \right| = 67.5^\circ$$

15. (c); In every 60 minutes, the minute hand gain 55 minutes on the hour hand.
16. (d); The needles will coincide each other at

$$= \frac{60}{11} \cdot 3 \text{ past } 3 = \frac{180}{11} \text{ past } 3$$

$$= 16 \frac{4}{11} \text{ past } 3$$

17. (b); The angle covered through 8 O' clock to 2 O' clock is 180° .
18. (a); The angle between minute hand and hour hand at 10.20 O' clock is 170° .
19. (a); 55 min spaces are covered in 60 min

$$60 \text{ min spaces are covered in } \frac{60}{55} \times 60 \text{ min}$$

$$= 65 \frac{5}{11} \text{ min}$$

$$\text{Loss in 64 min} = 65 + \frac{5}{11} - 64 = \frac{16}{11}$$

$$\text{Loss in 24 hrs} = \frac{16}{11} \times \frac{1}{64} \times 24 \times 60 = 32 \frac{8}{11} \text{ min}$$

20. (c); Given year is a leap year and every leap year calender repeat after 28 year so
2016 + 28 = 2044.
The calender of 2044 is similar as 2016 calender.

21. (b); Since 3rd day of the month = Tuesday.
So, 1st day of the month = Sunday
Therefore, 8th, 15th, 22th and 29th day of the month will be Sunday.
So, maximum 5 Sunday will be in a given month.

22. (b); 2nd August → Friday

12 August + 13 → 25 August
on 25 August, the day will be Sunday.

23. (c); In 30 days of month, day which is on 1st day or 2nd day will five times in a given month. Obviously it may be Friday or Saturday.

24. (c); Three Sunday are on even no. will be possible only if first Sunday is 2nd day of a month.
So, Sunday as in 2, 9, 16, 23, 30

25. (a); Three days after Wednesday = Saturday
Since 3rd January is Sunday
So, 2nd January will be Saturday
So, next Saturday will be 9, 16, 23, 30 January
∴ Required date = 30 January

26. (b); 3rd day → Monday
5th day → Wednesday
12th day → Wednesday
19th day → Wednesday
Then 18th day will be Tuesday.

27. (a); We know that in December, there are 31 days. So, day which are in 1st, 2nd and 3rd december will be maximum day.
Since 17th day of the month is Monday.
So, Monday date = 3, 10, 17, 24, 31 = 5 times

28. (c); 25th Oct. → Thursday
26th Oct. → Friday
27th Oct. → Saturday
28th Oct. → Sunday
29th Oct. → Monday

29. (b); Clock makes two right angle in an hour but it makes one right angle between 3 to 4 and 9 to 10, So in 24 hours, clock will make 44 times right angle.

30. (c); Saturday
World Environment day - 5 June
Children day - 14 Nov.
days between 5 June to 14 Nov.
June = 25
July = 31
Aug. = 31
Sep. = 30
Oct. = 31
Nov. = 14
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So, no. of odd day = 1

∴ Required day = Friday + 1 = Saturday